



MATHEMATICS LEARNING OUTCOMES FOR PRIMARY SCHOOLS, BY CLASS LEVELS AND BY TEACHING CYCLES

INFANT 1		
Cycle 1	Numbers & Number Operations	Number A MA 1.01 Identify and count groups of objects through oral exercises such as playing games, singing songs, and saying rhymes, initially to 10 and then beyond, using the counting principles of stable order, one-to-one correspondence, and cardinality. MA 1.02 Arrange, match, and create groups of up to ten objects to written numerical symbols.
		Number B MA 1.03 Count groups of objects, initially to 10 and then beyond, using the counting principles of abstraction and order irrelevance. MA 1.04 Identify and state how many objects are in a group of up to 10 objects at a glance without having to count them one by one. MA 1.05 Identify an individual number, a sequence of numbers and the number before, after or between given numbers using a number line. MA 1.06 Identify and recite numbers 1 to 30 in sequence with fluency and accuracy.
		Number C MA 1.07 Compare numbers from 0 to 10 using the less than, greater than and equals signs. MA 1.08 Identify and write the numeric symbols for numbers from 0 to 10. MA 1.09 Compose and decompose numbers from 1 - 10, grouping items into given numbers with no remainder. MA 1.10 Identify the position of an item in a group using ordinal numbers from first to tenth.
Cycle 2	Spatial Relationships & Shapes	Geometry A MA 6.01 Find and describe examples of points, lines, squares, circles, rectangles and triangles in the classroom, school and the wider environment. MA 6.02 Explore, classify and compare common shapes through play and use of manipulatives. MA 6.03 Identify and describe the properties of triangles, squares and rectangles in terms of number of sides and corners. MA 6.04 Construct and describe 2-D shapes using straws, sticks, clay, building blocks and other materials.
		Numbers & Number Operations Patterns MA 2.01 Identify and discuss examples of patterns in the classroom, school and wider environment. MA 2.02 Find and sort objects and shapes based on their color, size, number of sides and other attributes. MA 2.03 Create and describe patterns using objects, actions, shapes, colours, sounds, or numbers. MA 2.04 Group 10 or fewer objects into sets of 2's, 3's, 4's, and 5's without remainders MA 2.05 Count objects, initially to 10 and then beyond by 1s and 2s, forwards and backwards.
		Measurement A MA 7.01 Compare and discuss the length, height, mass, temperature and capacity of two objects using words such as longer, taller, shorter, lighter, heavier, colder, hotter, more full or emptier.



INFANT 1		
		MA 7.02 Investigate and determine the length of objects found in the classroom or wider environment, in non-standard units of measurement, such as finger lengths, pieces of string or lengths of a pencil. MA 7.03 Describe and compare the position of two objects relative to each other in terms of distance and direction using phrases such as near to, far from, very far from, to the right of, below, above and so on.
Cycle 3	Spatial Relationships & Shapes	Measurement B-Time MA7.04 Describe and give examples of the passage of time using terms such as minute, hour, day, week, month and year in everyday activities. MA 7.05 List and arrange in sequence the days of the week and months of year using ordinal numbers. MA 7.06 Identify and sequence the current dates and days of the month on a calendar or weather chart using ordinal numbers. MA 7.07 Tell, demonstrate and interpret time to the hour using an analogue clock.
	Numbers & Number Operations	Addition and Subtraction MA 3.01 Add and subtract sets of up to ten objects including with the use of zero MA 3.02 Explore and demonstrate strategies to add and subtract sets of up to ten objects with and without the use of concrete objects. MA 3.03 Solve problems involving the adding and subtracting up to 10 objects, using real-life situations.
		Fractions MA 5.01 Explain and demonstrate how a whole object can be divided into parts of equal and different sizes. MA 5.02 Describe and identify fractions in everyday situations by using language such as ‘1 out of 2’. MA 5.03 Compose and decompose a region, shape or set of objects using halves and quarters, showing that the fractional parts are equal. MA 5.04 Identify, compare and match halves and quarters and objects in parts with the symbols $\frac{1}{2}$ and $\frac{1}{4}$ using real-life situations.
Cycle 4	Spatial Relationships & Shapes	Geometry B MA 6.05 Identify and discuss planes in the classroom, school and wider environment. MA 6.06 Find, describe and compare examples of 3-D objects such as spheres, cubes, cylinders and cones in the classroom, school and the wider environment.
	Data Handling	Data MA 9.01 Gather and compile data from the environment through observation, counting, sorting and grouping of items such as objects and pictures. MA 9.02 Organize and display data using concrete materials in tally charts and on pictorial representations. MA 9.03 Create, display and interpret information presented in pictographs using a variety of data sets.



INFANT 2		
Cycle 1	Numbers & Number Operations	Number A MA 1.11 Identify and count numbers up to 100 using a number and the five counting principles of stable order, one to one correspondence, order irrelevance, cardinality and abstraction. MA 1.12 Identify and match number names with numeric symbols for numbers from 0-100 both orally and in writing. MA 1.13 Apply the concept of zero to real- life situations. MA 1.14 Identify the position and organize items in a group using ordinal numbers from first to one hundredth.
		Number B MA 1.15 Compose and decompose 2-digit numbers to form groups of tens and ones. MA 1.16 Sequence a set of numbers between 0 and 100 in ascending or descending order using a number line. MA 1.17 Identify the number that is ten more or ten less than a given number using a place value chart and apply it to real life situations. MA 1.18 Compare numbers from 0 to 100 using the less than, greater than and equal signs.
	Spatial Relationships & Shapes	Geometry A MA 6.07 Differentiate between horizontal, vertical and diagonal lines and draw rays and angles. MA 6.08 Identify the similarities and differences between triangles, squares, rectangles and circles. MA 6.09 Create by drawing or modelling and describe 2-D shapes with a specified number of sides using manipulatives.
Cycle 2	Numbers and Number Operations	Patterns MA 2.06 Identify and describe patterns in pictures and artistic designs. MA 2.07 Create and describe repeated patterns using actions, objects, colours, sounds, shapes, letters and numbers.
	Spatial Relationships & Shapes	Measurement A MA 7.08 Investigate and compare the perimeter and area of 2-D shapes using non-standard measures. MA 7.09 Measure and compare the length of lines, perimeter of shapes and real objects found in the environment using customary units of feet and inches. MA 7.10 Investigate and discuss the volume of 3-D shapes using non-standard units of measurement. MA 7.11 Measure and compare the volume of containers using the customary units of cups and pints and the mass of objects using customary units of pounds and ounces.
	Numbers & Number Operations	Addition and Subtraction A MA 3.04 Add a 1-digit number to a 2-digit number that ends in a zero and apply to real life situations. MA 3.05 Subtract a 1-digit number from a 2-digit number without the need to borrow and apply to real life situations. MA 3.06 Add a 1-digit number to any 2-digit number with the answer not exceeding 99 and without regrouping.



INFANT 2		
Cycle 3	Spatial Relationships & Shapes	Measurement B: Time MA 7.12 Identify and tell the time as half hour, quarter hour to or past the hour using an analogue clock. MA 7.13 Identify, explain and apply the terms a.m. and p.m. to time in real life situations.
	Numbers & Number Operations	Addition and Subtraction B MA 3.07 Add a single or two 2-digit numbers together with the answer not exceeding 100, vertically and horizontally with or without the place value chart and complete number sentence with sums up to 100 using +, =. MA 3.08 Subtract a single or 2-digit number from a 2-digit number, vertically and horizontally, without the need to barrow, with or without the place value chart and complete number sentence with differences up to 100 using -, =. MA 3.09 Explore and solve problems using the additive identity property.
		Fractions and Decimals MA 5.05 Compose and decompose a region, shape or set of objects using halves, thirds, quarters and fifths. MA 5.06 Identify and match fractional parts with the symbols $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$.
Cycle 4	Numbers & Number Operations	Multiplication and Division MA 4.01 Place and divide up to 50 objects or pictures into groups of 2's, 3's, 5's and 10's of equal size. MA 4.02 Investigate that multiplication is the same as repeated addition. MA 4.03 Multiply two 1-digit numbers together using manipulatives arranged in groups, multiplication arrays and so on.
	Spatial Relationships & Shapes	Geometry B MA 6.10 Identify, describe and compare 2-D shapes according to specific properties including length of sides and number of vertices through discussion and demonstrations to insert in a table. MA 6.11 Create and name models of 3-D shapes such as the cube, cuboid, sphere, cylinder, and cone or objects with specified properties, such as number of faces, edges and vertices through discussions and role-playing. MA 6.12 Investigate and explain the similarities and differences between symmetrical shapes.
	Data Handling	Data MA 9.04 Collect and organize data from pictures, written sources and the environment through observation. MA 9.05 Organize and display data using concrete materials in tally charts and column representations. MA 9.06 Collect and Interpret information presented in simple column graphs using a variety of datasets.



STANDARD 1

Cycle 1	Numbers & Number Operations	<i>Numbers A</i> MA 1.19 Read, write and match numbers up to 1000 using numerical symbols and words and apply to real life situations. MA 1.20 Identify and explain the value of each column in a place value chart as ten times more or less than the neighbouring column for numbers between 0 and 999. MA 1.21 Compare numbers up to 1000 using the symbols for equals (=), less than (<) and greater than (>) and apply to real life situations. MA 1.22 State, read and write numbers in expanded form, up to 1000. <i>Patterns</i> MA 2.08 Sequence and identify the next, or a missing non-consecutive number between 0 and 1000 in ascending and descending order, using a number line. MA 2.09 Create and describe increasing, decreasing, and alternating patterns using numbers, objects, actions, shapes, colours, or sounds. MA 2.10 Count forward and backward by 2's, 5's, 10's and 100's from any given starting number between 0 and 1000.
	Spatial Relationships & Shapes	<i>Geometry A</i> MA 6.11 Create and describe horizontal, vertical, diagonal, intersecting, parallel and perpendicular lines. MA 6.12 Investigate and discuss how the perimeter of common shapes such as triangles, squares and rectangles are calculated. MA 6.13 Create compound shapes using manipulatives such as pattern blocks, sticks, straws, string or other materials. MA 6.14 Describe the features of 3-D shapes such as cones, cylinders, cubes, cuboids and pyramids.
Cycle 2	Numbers & Number Operations	<i>Addition and Subtraction A</i> MA 3.10 Add & subtract 2-digit numbers without regrouping using a range of mental and written strategies. MA 3.11 Add and subtract 2-digit numbers with regrouping using manipulatives such as base ten blocks or Legos. MA 3.12 Add three 2-digit numbers with and without regrouping in unit columns.
	Spatial Relationships & Shapes	<i>Measurement A</i> MA 7.14 Estimate, measure, compare and record the length of lines, distances and the size of objects using the customary unit of inches, feet and yards. MA 7.15 Estimate, measure, compare and record the mass of various objects in the customary unit of pounds and ounces. MA 7.16 Estimate, measure compare and record the capacity of a container using the customary unit of cups, pints, quarts and gallons.
	Numbers & Number Operations	<i>Addition and Subtraction B</i> MA 3.13 Demonstrate and explain the relationship between addition and subtraction. MA 3.14 Add & subtract two 3-digit numbers without regrouping using unit columns. MA 3.15 Add & Subtract two 3-digits with regrouping using manipulatives such as base ten blocks.



STANDARD 1

Cycle 3	Spatial Relationships & Shapes	Measurement B MA 7.17 Convert among units within the customary system of length, mass and capacity. MA 7.18 Convert a length of time between minutes and seconds. MA 7.19 Identify, measure and record the temperature of the environment, in either degrees Celsius or Fahrenheit.
	Numbers & Number Operations	Multiplication and Division A MA 4.04 Explore the multiplicative identity of a number, that is if you multiply a number by 1, the product is that original number and solve problems using the multiplicative identity property. MA 4.05 Represent and solve multiplication problems both horizontally and vertically MA 4.06 Multiply a 2-digit number by a 1-digit number MA 4.07 Round-off to the nearest ten to estimate when multiplying.
		Fraction and Decimals MA 5.07 Describe and illustrate parts of a whole or of a set using fractions with numerators other than one such as $\frac{2}{3}$, $\frac{3}{4}$, $\frac{2}{5}$, $\frac{5}{6}$, $\frac{4}{10}$ MA 5.08 Compare and sequence fractions with like denominators with the aid of manipulatives such as pictures, number line, fraction strips. MA 5.09 Add two or more proper fractions with like denominators. MA 5.10 Read and Convert fractions with tenths to decimals, for example $\frac{3}{10}$ is the same as 0.3 MA 5.11 Read, add and subtract numbers with one decimal place.
Cycle 4	Numbers & Number Operations	Multiplication and Division B MA 4.08 Investigate that division is the same as repeated subtraction. MA 4.09 Divide single and 2-digit numbers by 2, 3, 4, 5, 10, without remainders. MA 4.10 Identify and explain the relationship between multiplication and division. MA 4.11 Solve word problems with real life applications using multiplication and division.
	Data Handling	Data MA 9.07 Collect, analyse and represent data contained in a tally chart or frequency table in different forms like pictographs & bar graphs. MA 9.08 Identify and discuss situations that involve chance such as certain, impossible or equally likely events and investigate probability using tables and graphs.



STANDARD 2		
Cycle 1	Numbers & Number Operations	Numbers A MA 1.23 Identify and state the value of a digit based on its position in a number up to 6 digits. MA 1.24 Read, write and apply numbers up to 100,000 using numerical symbols and words to real life situations. MA 1.25 Sequence and compare a set of non-consecutive numbers in ascending and descending order up to 100,000 using the place value system. MA 1.26 State, read and write whole numbers up to 100,000 in expanded form. MA 1.27 Round whole numbers up to 100,000 to specific place values.
	Spatial Relationships & Shapes	Geometry A MA 6.15 Identify and draw 2-D shapes based on their attributes up to 10 sides. MA 6.16 Identify and draw lines of symmetry in plane figures. MA 6.17 Draw circles of various sizes using a compass and identify the centre, radius, diameter and circumference of the circles. MA 6.18 Construct common shapes such as triangles, squares and rectangles and calculate the perimeter by adding the lengths of all sides using metric units.
	Numbers & Number Operations	Addition and Subtraction MA 3.16 Add and subtract positive numbers up to 5-digits, with and without regrouping. MA 3.17 Develop and write number sentences using mixed operations of addition and subtraction to solve word problems. MA 3.18 Solve problems using the commutative property of addition.
Cycle 2	Data Handling	Data MA 9.09 Collect and represent data on a bar graph, pictograph and dot plot using real-life situation. MA 9.10 Determine the median, mode, and range for a given set of data with both odd and even number of elements. MA 9.11 Identify, determine and predict the probability that an event will happen in a real-life situation with a finite number of possible outcomes using the phrase "with a probability of x out of y".
	Numbers & Number Operations	Multiplication and Division A MA 4.12 Define and find multiples of whole numbers. MA 4.13 Multiply two 2-digit numbers with and without regrouping. MA 4.14 Multiply whole numbers up to 100 with one decimal place. MA 4.15 Discuss and explain why commutative property applies to multiplication.
		Fractions and Decimals A MA 5.12 Identify, explain and compare equivalent fractions using pictures, number line, fraction strips or other manipulatives. MA 5.13 Find and solve problems with fractions that are equivalent to another by multiplying both the numerator and the denominator by the same number. MA 5.14 Compare and sequence groups of proper fractions with like and unlike denominators.



STANDARD 2		
Cycle 3		MA 5.15 Add and subtract fractions with like denominators.
	Numbers & Number Operations	Patterns MA 2.11 Create and analyse 2-dimensional pattern using only pictures. MA 2.12 Identify and Explain the difference between odd, even numbers, prime and composite numbers and explore and identify patterns for triangular numbers. MA 2.13 Identify and explore pattern rules for given patterns such as missing element of a pattern and patterns for triangular numbers
	Spatial Relationships & Shapes	Measurement MA 7.20 Measure, Compare and record the length of lines and the size of objects using meters, centimeters and millimeters. MA 7.21 Discuss and compare the distances to and from various places using kilometers. MA 7.22 Measure, compare, and record the mass of various objects using kilograms and grams and capacity of a container using litre and millilitre. MA 7.23 Record measures of time using minutes, seconds and hours and convert time from minutes to hours, hours to days and days to weeks.
Cycle 4	Numbers & Number Operations	Multiplication and Division B MA 4.16 Explore and discuss divisibility rules for division by 2, 5, and 10 by selecting numbers that follow each rule. MA 4.17 Divide 1-digit and 2-digit numbers by 2, 3, 4, 5, & 10, with and without remainders. MA 4.18 Divide 2-digit numbers by one-digit numbers using the short form of division without carrying over within the calculation.
	Numbers & Number Operations	Fraction and Decimals B MA 5.16 Add and subtract proper fractions with unlike denominators using area models. MA 5.17 Multiply and divide proper fractions with unlike denominators using area model. Fraction and Decimals C MA 5.18 Identify and write the value of any digit in a number that has up to 2 decimal places using the decimal place value chart. MA 5.19 State, read and write decimal numbers up to 2 decimal places in expanded form. MA 5.20 Compare, sequence and round-off numbers with 2 decimal places to the nearest tenth. MA 5.21 Identify the equivalent decimal forms of $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$. MA 5.22 Add and subtract decimal numbers, up to 2 decimal places.
	Spatial Relationships & Shapes	Geometry B MA 6.19 Identify the figure and construct 3-D figures from given nets. MA 6.20 Identify and describe turns using quarter, half, three-quarter and full turn and 0° , 90° , 180° , 270° , 360° . MA 6.21 Identify and classify an angle as acute, right, obtuse or straight based on the approximate size of the angle.



STANDARD 3		
Cycle 1	Numbers & Number Operations	Numbers A MA 1.28 Identify and state the value of a digit based on its position in a number up to 7 digits. MA 1.29 Round off very small and very large numbers using an appropriate method and apply to real life situations. MA 1.30 Read and write numbers using the Mayan and Roman Numeral numbering system. MA 1.31 Explore and apply square numbers to real life situations.
	Spatial Relationships & Shapes	Geometry A MA 6.22 Draw, measure and record the degrees of various angles and interior angles of various shapes using a protractor. MA 6.23 Identify and name a triangle as being acute, obtuse or right-angled. MA 6.24 Investigate the sum of the interior angles of triangles and quadrilaterals.
	Numbers & Number Operations	Addition and Subtraction MA 3.19 Add and subtract numbers up to 7 digits, with and without regrouping. MA 3.20 Solve problems and explain the associative property of addition.
Cycle 2	Numbers & Number Operations	Multiplication and Division A MA 4.19 Explain and solve problems using the associative property of multiplication. MA 4.20 List and demonstrate the squares of numbers up to 10, concretely and pictorially. MA 4.21 Discuss and find the lowest common multiples for a range whole number. MA 4.22 Multiply a 3-digit number by a 1-digit number or 2-digit number. MA 4.23 Multiply a whole number with a number with up to three decimal places.
		Multiplication and Division B MA 4.24 Explore and solve problems using divisibility rules for division by 3, 6, and 9. MA 4.25 Solve problems by dividing any double-digit number by a single-digit number. MA 4.26 Divide 2-digit, 3-digit and 4-digit numbers by 1-digit numbers, using the short form of division including carrying over within the calculation. MA 4.27 Divide a number with up to 2 decimal places by a single digit number using the short form of division. MA 4.28 Differentiate between Factors and Greatest Common Factors and find Factors and Greatest Common Factors of various numbers.
		Fraction and Decimals A MA 5.23 Reduce a proper fraction to its simplest form. MA 5.24 Compare and sequence a group containing both mixed numbers and improper fractions and convert improper fraction to mixed number. MA 5.25 Identify and justify the mixed number that is equivalent to a given decimal. MA 5.26 Add and subtract mixed numbers in real life situations.



STANDARD 3		
		MA 5.27 Multiply whole numbers and fractions by fractions.
Cycle 3	Numbers & Number Operations	Patterns MA 2.14 Investigate different ways to arrange a set of items to create a variety of patterns related to daily lives. MA 2.15 Solve problems involving the identification of missing elements in a pattern by investigating relationships between successive elements using addition, subtraction, multiplication and division.
	Spatial Relationships & Shapes	Measurement MA 7.24 Estimate using metric units, length, mass and capacity of a line or object. MA 7.25 Measure and record length, mass, capacity, and temperature, using metric units, including decimals. MA 7.26 Solve problems using metric units of length, mass, capacity and temperature in real life situations. MA 7.27 Convert between a 12-hour clock and a 24-hour clock and estimate the length of time an event takes in real life situations. MA 7.28 Measure the elapsed time of an event using a stopwatch to the nearest tenth of a second and calculate the elapsed time of an event that are many weeks, months or years apart.
	Data Handling	Data MA 9.12 Collect data and determine the median, mode, and range of the set of data with either an even or an odd number of elements. MA 9.13 Compute the mean for a set of numbers related to real-life situation and solve real-world problems involving median, mode and range. MA 9.14 Describe and predict outcomes from data using the language of chance or likelihood. MA 9.15 Collect and represent real-life data in bar graphs and histograms.
Cycle 4	Numbers & Number Operations	Fraction and Decimals B MA. 5.28 Identify the value of any digit in a number with up to 3 decimal places using a place value chart. MA 5.29 State, read and write decimal numbers up to 3 decimal places in usual and expanded form. MA 5.30 Compare, sequence and round off decimal numbers with up to 3 decimal places to the nearest tenth and hundredth. MA 5.31 Add and subtract decimals, up to 3 decimal places. MA 5.32 Converting fractions that have 2, 4, 5, 8, 10, 20, 25 or 50 as the denominator to decimal numbers.
	Data Handling	Sets MA 8.01 Create and describe sets based on the common features and attributes of numbers, people, objects and other entities. MA 8.02 Identify elements that are not members of a set. MA 8.03 Investigate and empty (null) set.



STANDARD 3

	Spatial Relationships & Shapes	<i>Geometry B</i> MA 6.25 Calculate the perimeter of any 2-D shape and various compound shapes that have one or more missing measurements MA 6.26 Calculate the area of triangles, squares and rectangles using the formulas. MA 6.27 Create a design using tessellation
--	---	---



STANDARD 4		
Cycle 1	Numbers & Number Operations	Numbers A MA 1.32 Read, write and sequence positive and negative integers in ascending and descending order using a number line. MA 1.33 Construct a number line that shows both positive and negative integers.
	Spatial Relationships & Shapes	Geometry A MA 6.28 Identify and classify triangles as equilateral, isosceles, scalene, right-angle, acute and obtuse. MA 6.29 Draw and list properties of triangles with given angles and lengths of side using a ruler and protractor. MA 6.30 Investigate angles in triangles to deduce relationships between the interior and exterior angles. MA 6.31 Calculate area of a compound shape constructed from squares, rectangles and triangles.
	Numbers & Number Operations	Addition and Subtraction MA 3.21 Add or subtract a series of at least five numbers mentally. MA 3.22 Add and subtract positive numbers to a negative number using a number line. MA 3.23 Add and subtract a range of decimal numbers from the very small to the very large. MA 3.24 Identify and explain the difference between the commutative and associative properties of addition.
Cycle 2	Numbers & Number Operations	Multiplication and Division A MA 4.29 Multiply a positive number with a negative number. MA 4.30 Multiply two decimal numbers with up to 3 decimal places. MA 4.31 Explore how the use of brackets can change the order of operations in problems involving multiplication and addition or subtraction.
		Fraction and Decimals A MA 5.33 Explore and demonstrate the steps to find lowest common denominator of two or more fractions. MA 5.34 Add and subtract two or more fractions with unlike denominators using the lowest common denominator. MA 5.35 Multiply and divide mixed numbers. MA 5.36 Divide a whole number or a fraction by a fraction.
	Spatial Relationships & Shapes	Geometry B MA 6.32 Find a grid square on a map using coordinates and references. MA 6.33 Locate points and construct shapes and lines on the first quadrant of the coordinate graph using (x,y) coordinates.
Cycle 3	Numbers & Number Operations ,	Fraction and Decimals B MA 5.37 Identify common everyday situations where percent is used and represent various percentages using pictures, shaded areas, and fractional parts using real life situations. MA 5.38 Identify and investigate the relationship between percentage and fraction with a denominator of 100. MA 5.39 Discuss and describe real-life situations involving comparisons between percentages.



STANDARD 4

	Data Handling	Data MA 9.16 Collect and organize data in a frequency table and construct bar graph, histogram and line graphs and describe trends from line graph. MA 9.17 Construct and extract information from line and circle graphs to formulate solutions for real-life situations. MA 9.18 Collect data and find the mean, median, mode and range of a data sets of up to fifteen quantities.
	Spatial Relationships & Shapes	Measurement MA 7.29 Investigate the relationship between the metric and customary units of measurement by estimating and measuring the length of the same line or object using both systems. MA 7.30 Investigate the relationship between the metric and customary units of measurement by estimating and weighing the same object using both systems. MA 7.31 Investigate the relationship between the metric and customary units of capacity by estimating and finding the capacity of the same container for both systems.
Cycle 4	Numbers & Number Operations	Multiplication and Division B MA 4.32 Explore and demonstrate divisibility rules for division by 4, 7 and 8. MA 4.33 Divide a 2-digit number by a 1-digit number with or without remainders MA 4.34 Divide a whole number by a 2-digit number using long division. MA 4.35 Divide a whole number by a decimal between 0 and 1. MA 4.36 Explain how the use of brackets can change the order of operations in problems involving division and addition or subtraction.
	Data Handling	Sets MA 8.04 Define and list elements of finite and infinite sets. MA 8.05 Define and create equal sets with objects and different element. MA 8.06 Describe and create Venn diagram with the intersection of two (2) sets using of real world scenarios.
	Spatial Relationships & Shapes	Geometry MA 6.40 Construct and define the parts of a circles such as center, radius, diameter and circumference. MA 6.41 Illustrate how a circle is divided into 360 equal degrees. MA 6.42 Construct a circle divided into sectors with a given number of degrees using a compass and protractor.



STANDARD 5		
Cycle 1	Numbers & Number Operations	Numbers A MA 1.34 Read and write very large numbers in standard form, expanded form and scientific notation. MA 1.35 Round off a whole number to a specified number of significant figures and apply to real life situations. MA 1.36 Read, write and expand numbers with three or more decimal places. MA 1.37 Categorize and differentiate between odd, even, prime, and composite numbers. MA 1.38 Identify and illustrate the square of all integers between 1 and 20 and of the integers 25 and 100. MA 1.39 Demonstrate and illustrate the concept of cube numbers concretely and pictorially.
		Patterns MA 2.16 Represent and analyse patterns in real life data using tables, graphs, diagrams and manipulatives. MA 2.17 Investigate and solve arithmetic and geometric progressions using real-life data.
		Addition and Subtraction MA 3.25 Add and subtract, mentally numbers with up to three digits. MA 3.26 Add and subtract integers with like and unlike signs, using real life situations. MA 3.27 Add and subtract numbers with decimals to solve real world problems.
Cycle 2	Numbers & Number Operations	Multiplication and Division MA 4.37 Multiply and divide any two-digit number by any number between 1 and 20. MA 4.38 Multiply and divide a positive a number by a negative number. MA 4.39 Describe real life situations using the terms multiple and factor. MA 4.40 Identify the prime factor of 1-digit and 2-digit numbers and express natural numbers as products of their prime factors.
		Fractions and Decimals MA 5.40 Discuss and solve real world problems that require the adding and subtracting of mixed numbers. MA 5.41 Discuss and solve real world problems that require the multiplying and dividing of mixed numbers. MA 5.42 Discuss and solve real world problems with whole number, factions and decimals that involve more than one types of operation. MA 5.43 Compare a variety of decimals, fractions, and percent using greater than, less than or equal to and solve problems that involves finding the percentage increase or decrease of real-world quantities.
		Fractions and Decimal- Ratio MA 5.49 Express a ratio as part to part and part to whole and solve real-world problems involving ratios with 2 elements. MA 5.50 Explain the difference between ratios and rates and solve real-world problems involving ratio and rates.



STANDARD 5

Cycle 3	Spatial Relationships & Shapes	<i>Geometry A</i> MA 6.34 Research and solve real world problems involving the perimeter and area of triangles and quadrilaterals. MA 6.35 Draw circles and calculate the circumference, radius and diameter of circles using formulas. MA 6.36 Identify and construct cubes, cuboids, cylinders, cones, pyramids, and other 3-dimensional objects. MA 6.37 Investigate how to calculate the surface area and volume of cuboids.
		<i>Measurement</i> MA 7.32 Solve real-world problems that require conversion between units of measurement within the same system and between the customary and metric units of measurement. MA 7.34 Solve problems that include measurements expressed as negative numbers, such as temperature below zero and BCE. MA 7.35 Calculate speed based in actual real-world measurements of time and distance. MA 7.36 Solve problems in which the start time, end time and elapsed time is an unknown quantity.
		<i>Geometry B</i> MA 6.38 Plot a line on the first quadrant of a coordinate graph using given (x,y) coordinates and plot the results of a reflection on a rectangular grid. MA 6.39 Investigate how a variety of polygons can be found in tessellations of triangles and create tessellations from more than one type of polygons.
Cycle 4	Data Handling	<i>Statistics</i> MA 9.19 Construct circle graphs and line graphs from frequency tables using real-world data. MA 9.20 Calculate the mean, median, mode and range of large sets of real-world data.
		<i>Sets</i> MA 8.07 Identify and describe the elements in the union and intersection of two sets based on a Venn diagram using the set notation: { }, $\cap$, $\cup$. MA 8.08 Create subsets from universal sets and indicate the elements in the subsets using set notation.
		<i>Probability</i> MA 9.21 Explore probability and represent the outcome of a probability experiment using fractions and percents. MA 9.22 Represent the outcome of a probability experiment in a frequency table and on a bar graph. MA 9.23 Determine probability from data given in bar graphs, pictographs and circle graphs.



STANDARD 6		
Cycle 1	Numbers & Number Operations	Numbers A MA 1.40 Explain the value of digits in numbers smaller than 0.001 with the use of a place value chart and apply to real life situations. MA 1.41 Round off decimal numbers up to two significant figures and large numbers to up to three significant figures. MA 1.42 Categorize and differentiate between whole numbers, prime numbers, composite numbers and integers.
		Numbers B MA 1.43 Define the terms exponent and power to apply the concept of powers using repeated multiplication using 2 as the base number. MA 1.44 Identify and write the square root of a perfect square. MA 1.45 Identify and list the cubes of integers from 1 to 5 and of the number 10. MA 1.46 Convert to and from base 10 and base 2 using place value charts.
		Addition and Subtraction MA 3.28 Add and subtract positive and negative numbers to solve mental and real-world problems. MA 3.29 Solve and discuss real world word problems using addition and subtraction when there is an unknown quantity. MA 3.30 Compare the commutative and associative properties of addition and explain the importance of each using real world examples.
Cycle 2	Numbers & Number Operations	Multiplication and Division MA 4.41 Mentally multiply any 2-digit number by 4,5, 9, multiples of 10 and 25 & 50 using shortcuts. MA 4.42 Multiply and divide two negative numbers. MA 4.43 Identify the factors of negative whole numbers. MA 4.44 Solve real-world multiplication and division problems to solve word problems in which there is an unknown quantity. MA 4.45 Explain the importance of the commutative property of multiplication using real world examples. MA 4.46 Solve problems requiring application of order of operations.
		Fractions and Decimals A MA 5.44 Convert decimals to fractions for any value between 0 and 1 and vice versa. MA 5.45 Add, subtract, multiply and divide different types of fractions using real-life scenarios.
Cycle 3	Spatial Relationships & Shapes	Geometry A MA 6.40 Determine the perimeter of a 2D shape when the length of one or more sides is not known. MA 6.41 Investigate and calculate the area of quadrilaterals and compound shapes MA 6.42 Investigate and calculate surface area of cylinders and pyramids. MA 6.43 Investigate and calculate the volume of cylinders, rectangular pyramids and cones using formulas.



STANDARD 6

		<i>Geometry B</i> MA 6.44 Construct regular polygons using ruler, protractor and compass and investigate the interior and exterior angles. MA 6.45 Identify figures that are similar and congruent. MA 6.46 Plot on a rectangular grid the results of rotation.
	Data Handling	<i>Sets</i> MA 8.09 Describe sets using the terms and appropriate notation for set, subset, element, empty set, null set, universal set, intersection and union. MA 8.10 Create and interpret Venn diagrams of up to three sets from real-world data.
Cycle 4	Data Handling	<i>Data A</i> MA 9.24 Create a line graph, bar graph and a histogram based on two or more data sets and make predictions after analysis MA 9.25 Use a table of values to plot a line on a coordinate graph. <i>Statistics</i> MA 9.26 Analyze, explain and solve real world problems of mean, median and mode. <i>Probability</i> MA 9.27 Explain that probability is a measure on a scale of 0-1 of how likely an event is to occur. MA 9.28 Determine the probability of a sequence of events using tree diagrams. MA 9.28 Determine the probability of an event from various types of charts, graphs and tables. MA 9.30 Explore the probabilities involved with games of chance. MA 9.31 Represent the outcome of a probability experiment on a circle graph.